Patent-Style Technical Drawings

TempSim – one-dimensional transient heat conduction (Temperatur1D)

Seven drawings of the TempSim subject matter, each rendered in the two customary patent drawing styles:

- Numbered – black line art with reference numerals and curved lead lines; the numerals are explained in the reference list on each page.
- Labelled – the same view with direct text labels and straight leader lines.

Hatching marks sectioned parts, dash-dot lines are centre lines, dashed lines are hidden edges or cell faces. Every drawing is also stored as a separate vector PDF and PNG under the file name given below (folder docs/patent_drawings/).

Index of drawings

FIG.	Style	File name (.pdf / .png)	Page
1	**Wall layer discretised into nodes (cross-section)**		
	reference numerals	`fig01_wall_discretisation_numbered.pdf`	2
	direct labelling	`fig01_wall_discretisation_labelled.pdf`	3
2	**Laboratory test rig, side elevation partly in section**		
	reference numerals	`fig02_test_rig_numbered.pdf`	4
	direct labelling	`fig02_test_rig_labelled.pdf`	5
3	**Block diagram of the simulation system**		
	reference numerals	`fig03_block_diagram_numbered.pdf`	6
	direct labelling	`fig03_block_diagram_labelled.pdf`	7
4	**Flow chart of the simulation method**		
	reference numerals	`fig04_flow_chart_numbered.pdf`	8
	direct labelling	`fig04_flow_chart_labelled.pdf`	9
5	**Temperature–time diagram of the nodes**		
	reference numerals	`fig05_temperature_diagram_numbered.pdf`	10
	direct labelling	`fig05_temperature_diagram_labelled.pdf`	11
6	**Detail of one control volume with energy balance**		
	reference numerals	`fig06_control_volume_numbered.pdf`	12
	direct labelling	`fig06_control_volume_labelled.pdf`	13
7	**Graphical user interface of the program**		
	reference numerals	`fig07_user_interface_numbered.pdf`	14
	direct labelling	`fig07_user_interface_labelled.pdf`	15

Summary file: patent_drawings_summary.pdf · generated 2026-11-02 by make_drawings.py

FIG. 1 – Wall layer discretised into nodes (cross-section)

File: fig01_wall_discretisation_numbered.pdf (+ .png) · Style: reference numerals

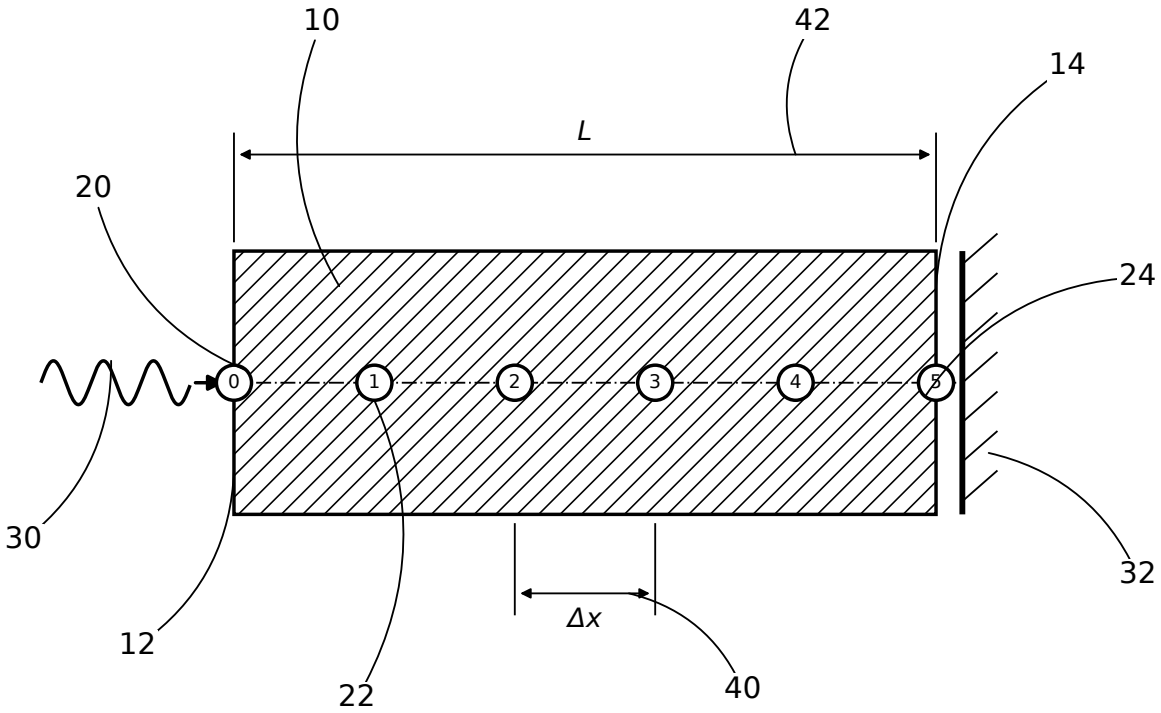


FIG. 1

Reference list

10	wall layer	**24**	boundary node i = 5
12	first (exposed) surface	**30**	time-varying temperature source $T_0(t)$
14	second surface	**32**	constant-temperature boundary T_5
20	boundary node i = 0	**40**	node spacing
22	interior node i = 1…4	**42**	wall thickness

FIG. 1 – Wall layer discretised into nodes (cross-section)

File: fig01_wall_discretisation_labelled.pdf (+ .png) · Style: direct labelling

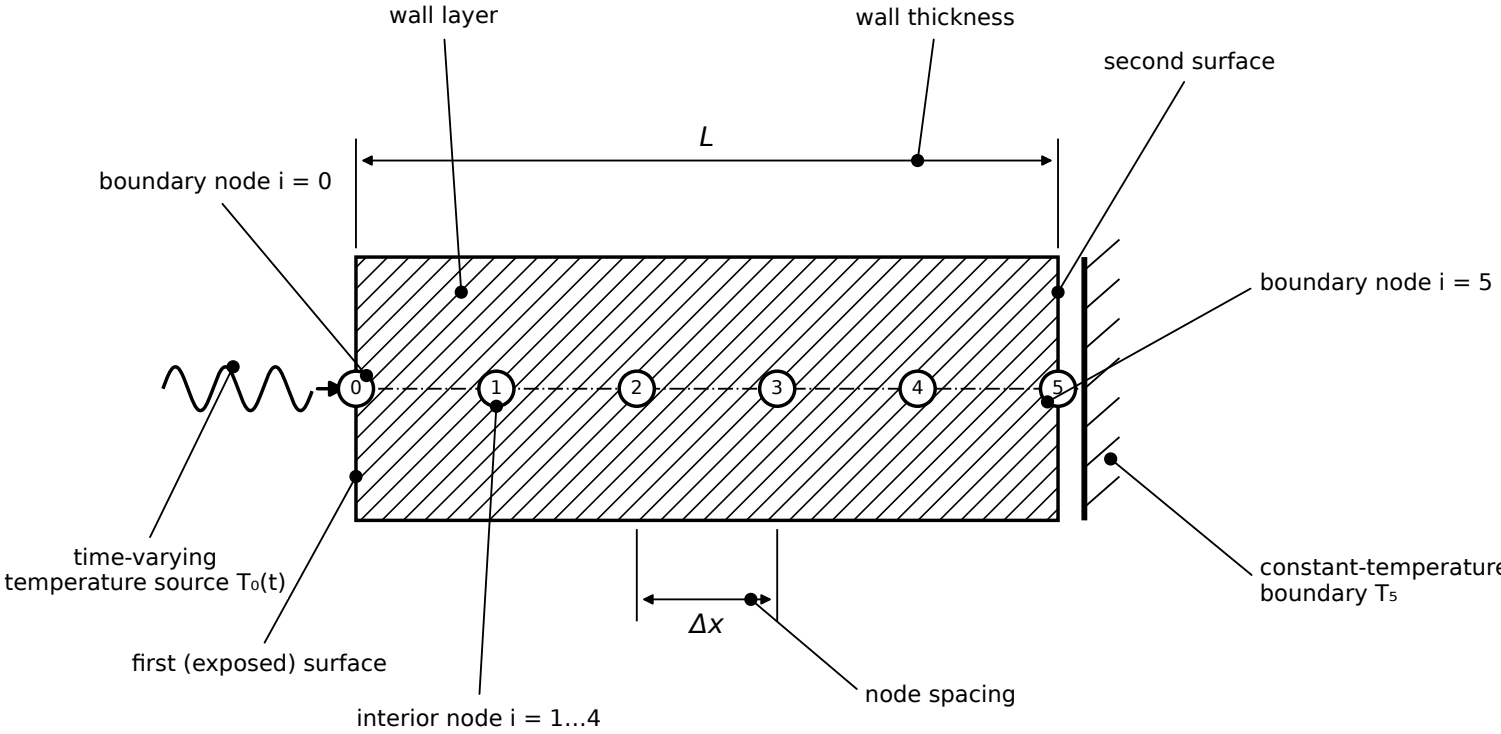


FIG. 1

Direct-labelling variant of the drawing on the previous page: same view and part layout, with the reference numerals replaced by plain-text labels on straight leader lines.

FIG. 2 – Laboratory test rig, side elevation partly in section

File: fig02_test_rig_numbered.pdf (+ .png) · Style: reference numerals

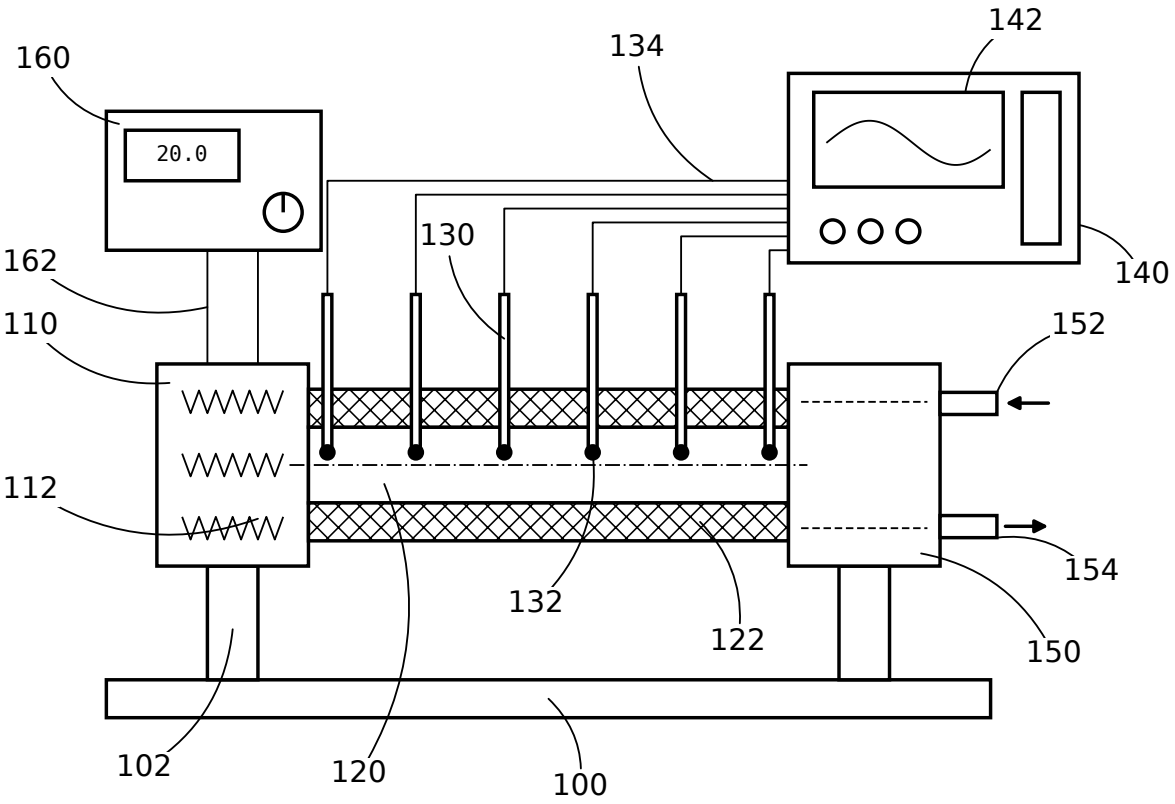


FIG. 2

Reference list

100	base plate	**134**	signal cable
102	support	**140**	data logger
110	heater block	**142**	display
112	heating element	**150**	cooling block
120	specimen rod	**152**	coolant inlet
122	thermal insulation	**154**	coolant outlet
130	thermocouple	**160**	temperature controller
132	measuring junction	**162**	power lines

FIG. 2 – Laboratory test rig, side elevation partly in section

File: fig02_test_rig_labelled.pdf (+ .png) · Style: direct labelling

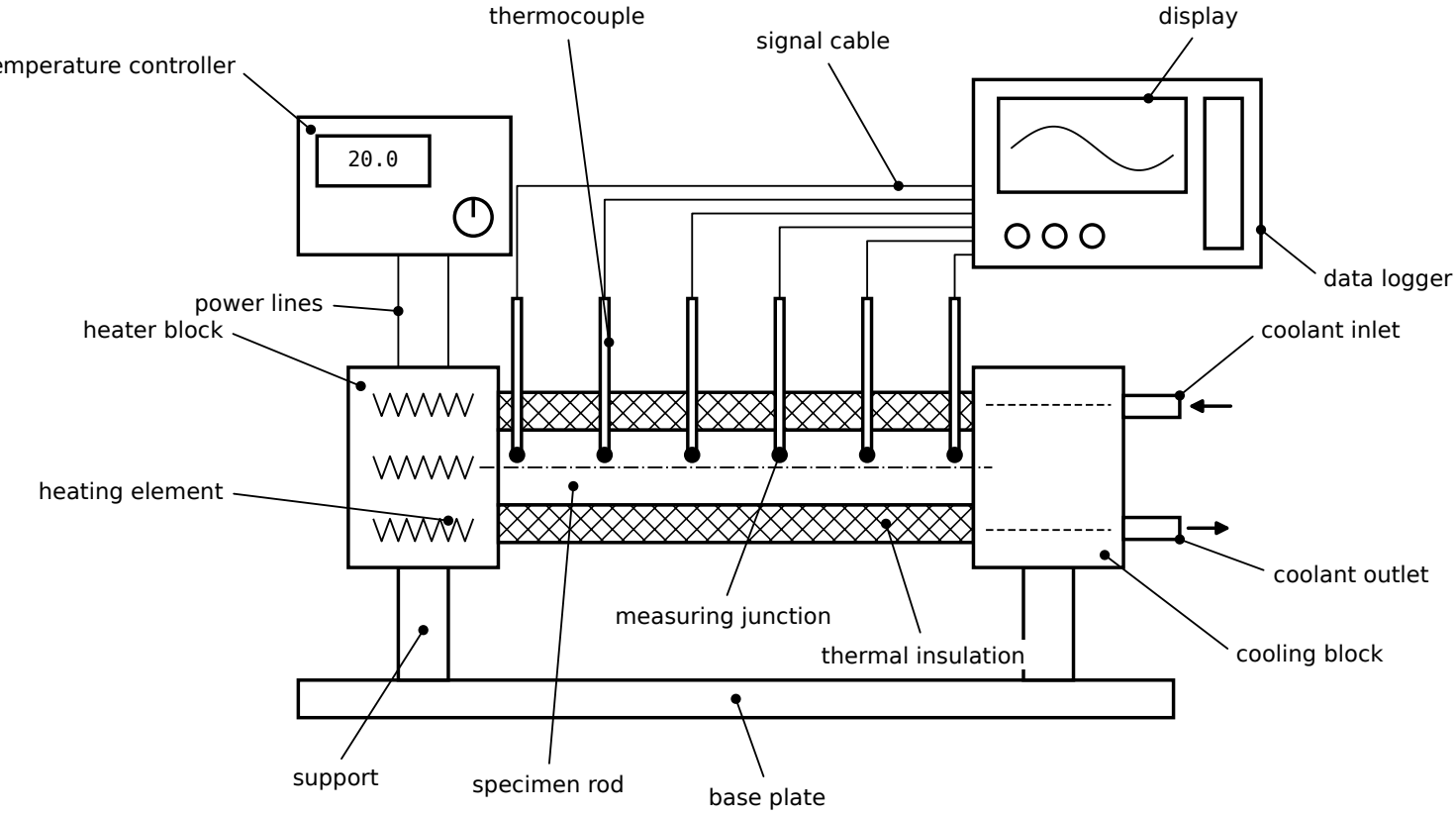


FIG. 2

Direct-labelling variant of the drawing on the previous page: same view and part layout, with the reference numerals replaced by plain-text labels on straight leader lines.

FIG. 3 – Block diagram of the simulation system

File: fig03_block_diagram_numbered.pdf (+ .png) · Style: reference numerals

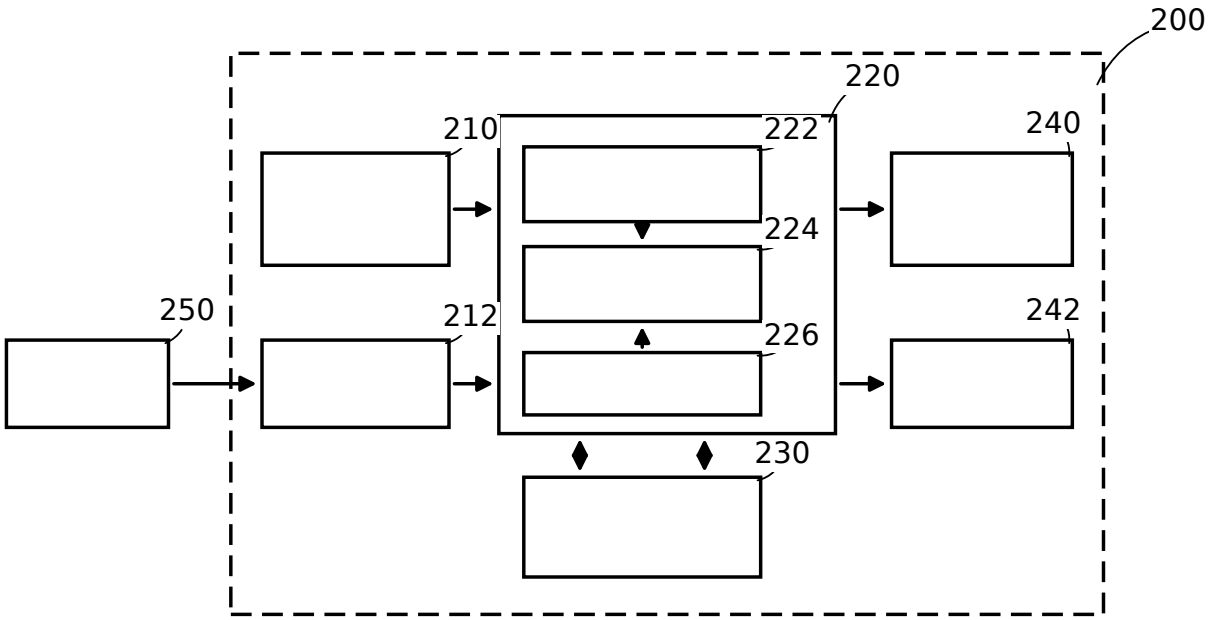


FIG. 3

Reference list

200	simulation system	**226**	time-step controller
210	input unit	**230**	memory (temperature arrays)
212	sensor interface	**240**	display unit
220	processor	**242**	data export
222	boundary-condition module	**250**	temperature sensors
224	explicit finite-difference solver		

FIG. 3 – Block diagram of the simulation system

File: fig03_block_diagram_labelled.pdf (+ .png) · Style: direct labelling

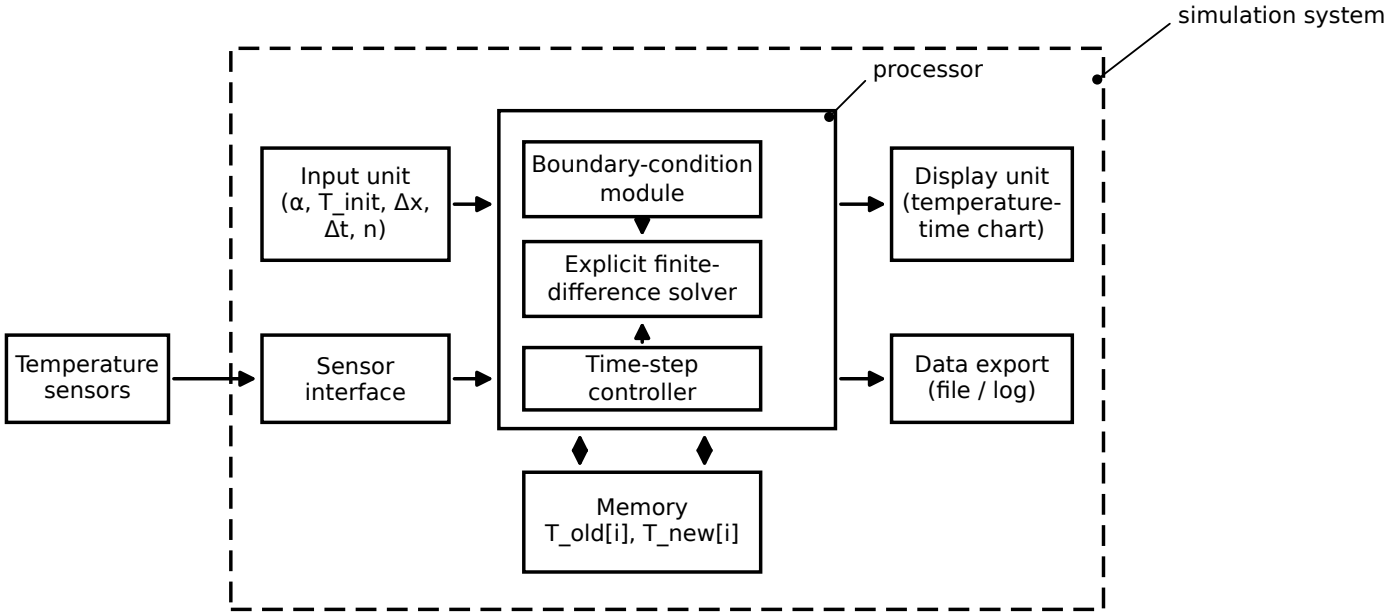


FIG. 3

Direct-labelling variant of the drawing on the previous page: same view and part layout, with the reference numerals replaced by plain-text labels on straight leader lines.

FIG. 4 – Flow chart of the simulation method

File: fig04_flow_chart_numbered.pdf (+ .png) · Style: reference numerals

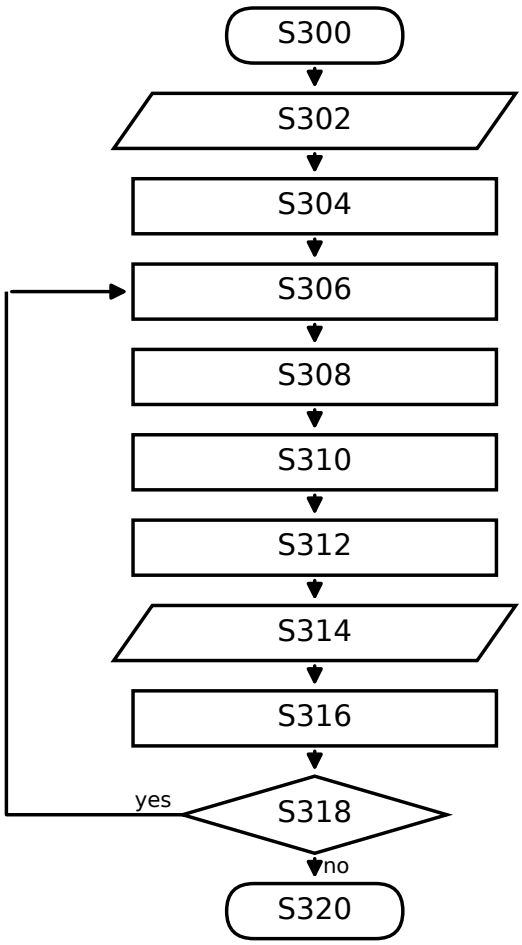


FIG. 4

Reference list

S300 Start

S302 Read parameters α, T_init, Δx, Δt, n

S304 Initialise all nodes T_i = T_init, t = 0

S306 Advance time t = t + Δt

S308 Set boundary node T_0 = T_init + 5·cos(π t / 3600 s)

S310 Update interior nodes i = 1…4 T_i' = T_i + αΔt/Δx²·(T_{i-1} − 2T_i + T_{i+1})

S312 Hold far boundary T_5 = T_init

S314 Output T_i(t) to chart

S316 Copy T' → T

S318 Step < n ?

S320 End

FIG. 4 – Flow chart of the simulation method

File: fig04_flow_chart_labelled.pdf (+ .png) · Style: direct labelling

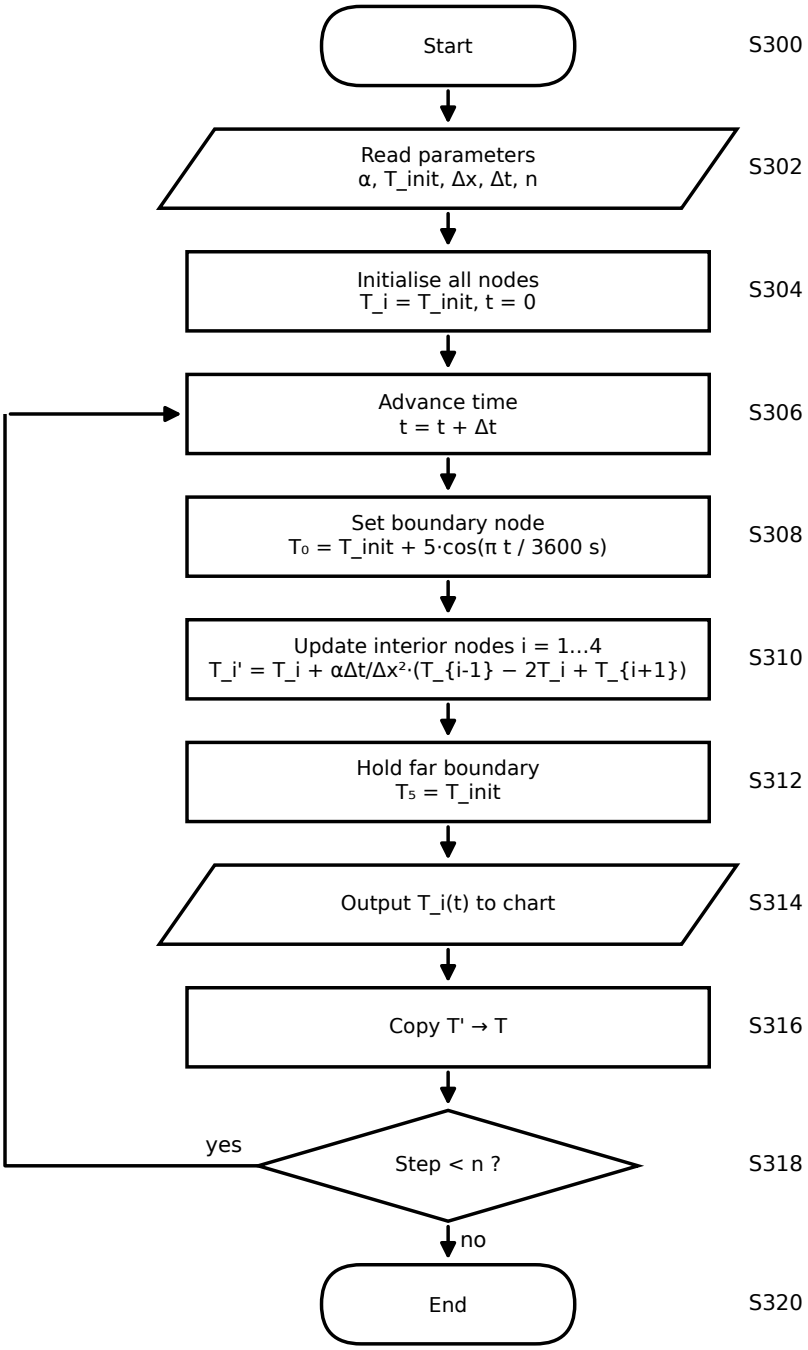


FIG. 4

Direct-labelling variant of the drawing on the previous page: same view and part layout, with the reference numerals replaced by plain-text labels on straight leader lines.

FIG. 5 – Temperature–time diagram of the nodes

File: fig05_temperature_diagram_numbered.pdf (+ .png) · Style: reference numerals

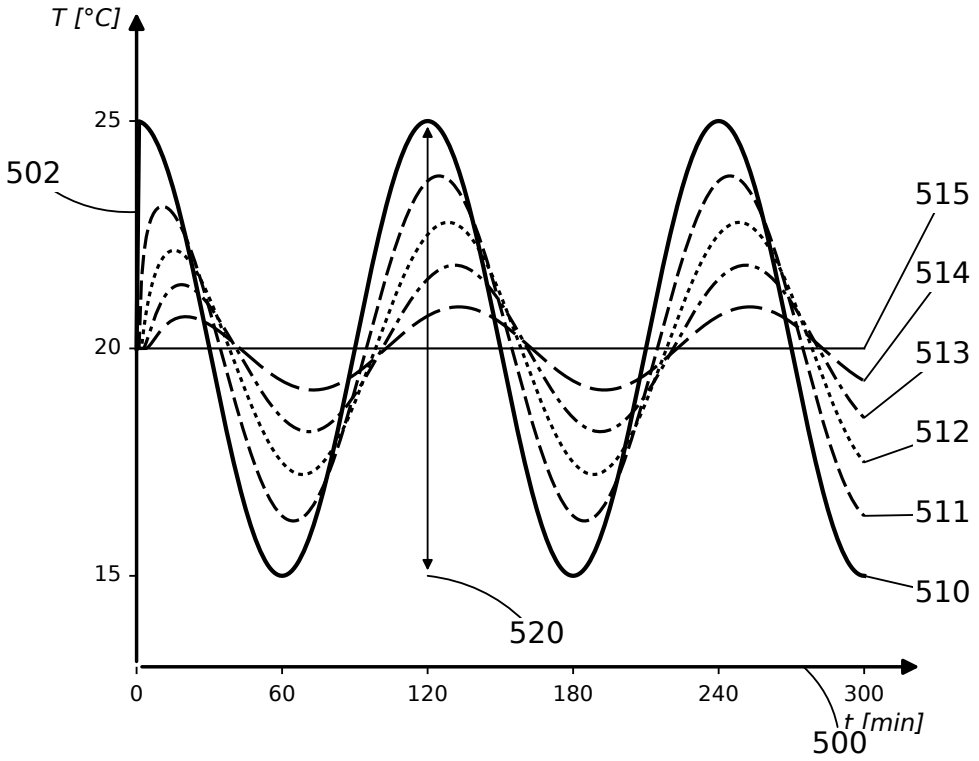


FIG. 5

Reference list

500	time axis	**513**	node 3
502	temperature axis	**514**	node 4
510	node 0 – excited surface	**515**	node 5 – constant boundary
511	node 1	**520**	temperature amplitude ±5 K
512	node 2		

FIG. 5 – Temperature–time diagram of the nodes

File: fig05_temperature_diagram_labelled.pdf (+ .png) · Style: direct labelling

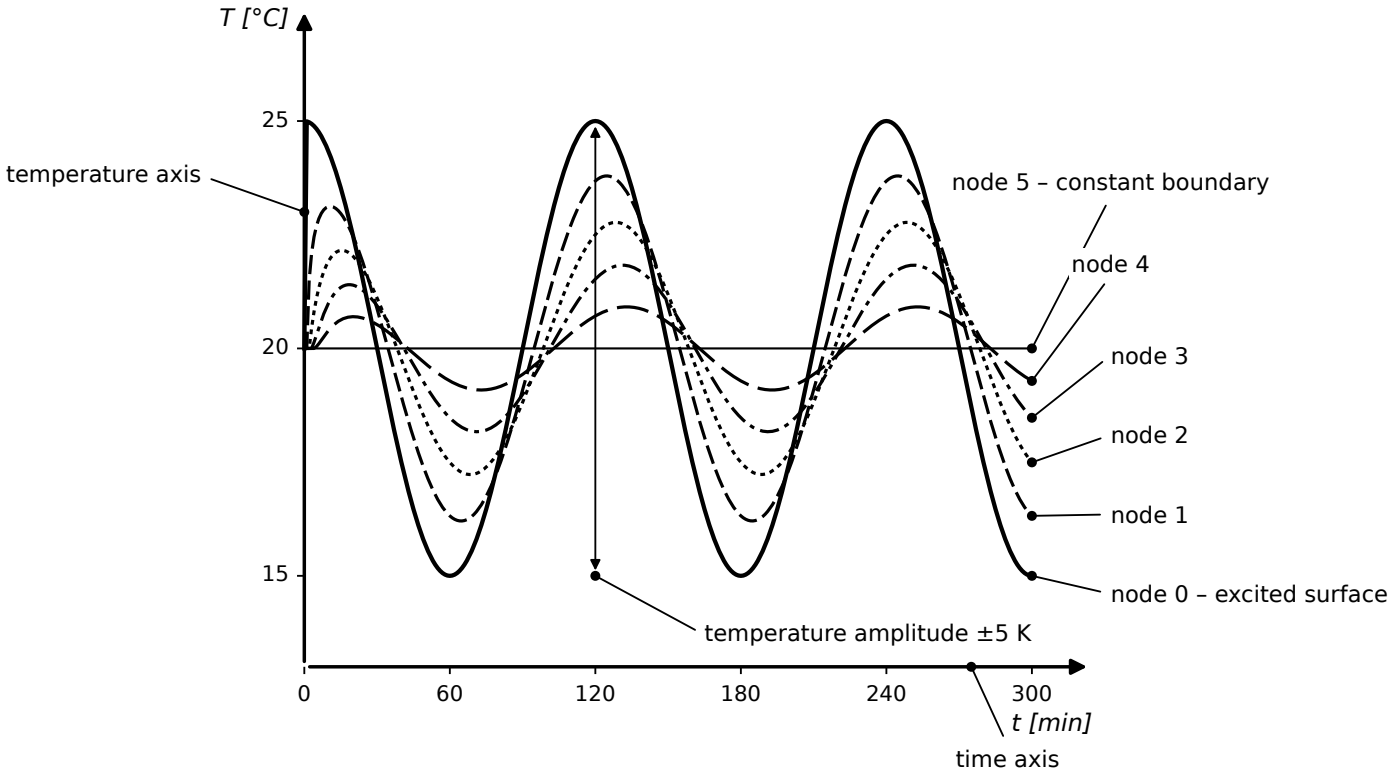


FIG. 5

Direct-labelling variant of the drawing on the previous page: same view and part layout, with the reference numerals replaced by plain-text labels on straight leader lines.

FIG. 6 – Detail of one control volume with energy balance

File: fig06_control_volume_numbered.pdf (+ .png) · Style: reference numerals

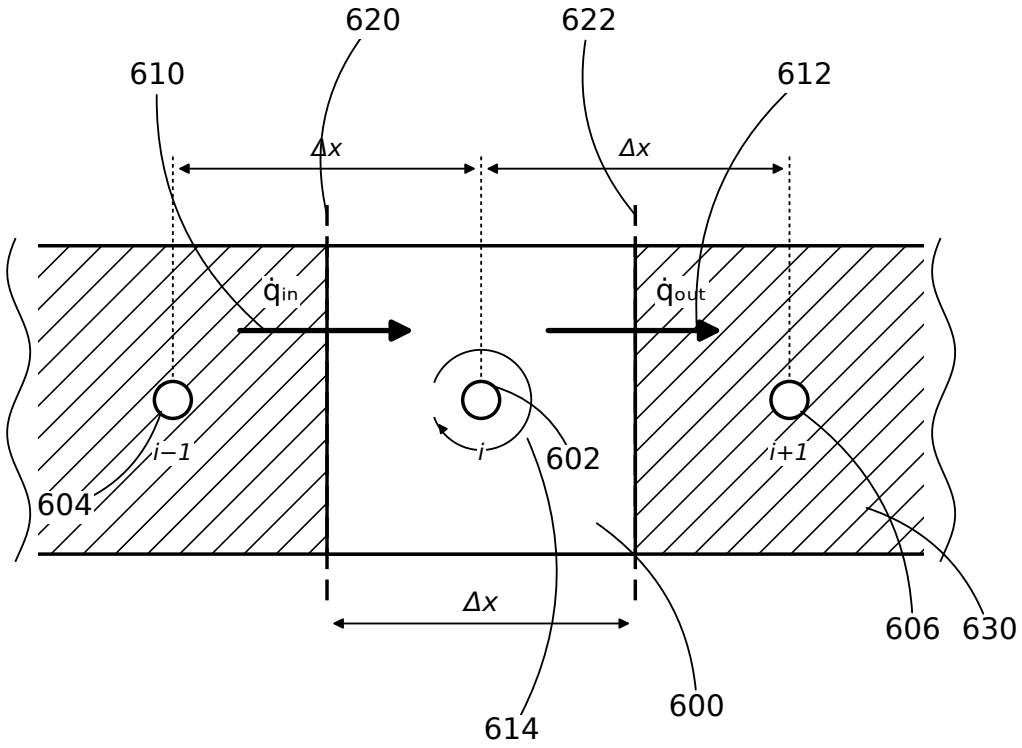


FIG. 6

Reference list

600	control volume	**612**	outgoing heat flux
602	node i	**614**	stored energy ρc·Δx·dT/dt
604	neighbour node i−1	**620**	cell face i−½
606	neighbour node i+1	**622**	cell face i+½
610	incoming heat flux	**630**	adjacent solid material

FIG. 6 – Detail of one control volume with energy balance

File: fig06_control_volume_labelled.pdf (+ .png) · Style: direct labelling

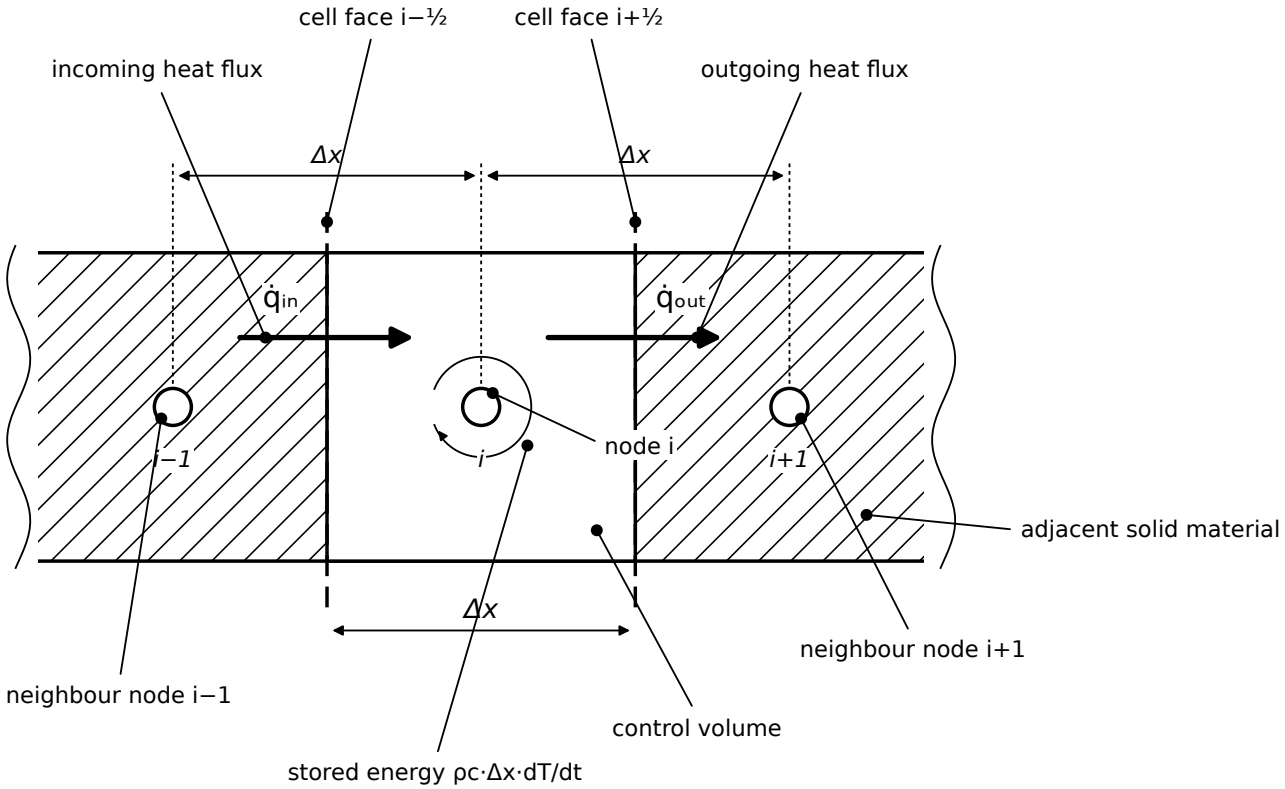


FIG. 6

Direct-labelling variant of the drawing on the previous page: same view and part layout, with the reference numerals replaced by plain-text labels on straight leader lines.

FIG. 7 – Graphical user interface of the program

File: fig07_user_interface_numbered.pdf (+ .png) · Style: reference numerals

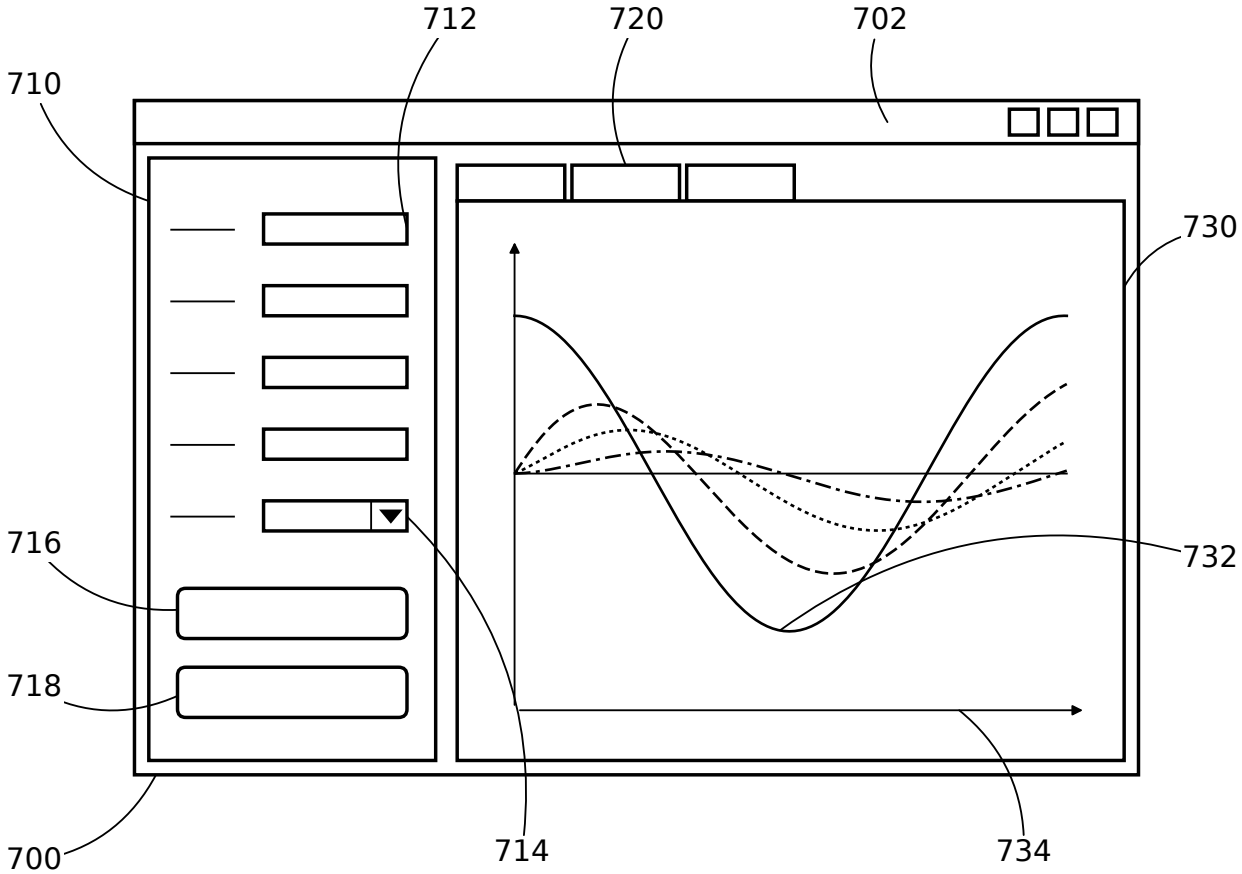


FIG. 7

Reference list

700	program window	**718**	single-step button
702	title bar	**720**	tab control
710	parameter panel	**730**	chart area
712	numeric input fields	**732**	node temperature curves
714	diffusivity selector	**734**	time axis
716	start button		

FIG. 7 – Graphical user interface of the program

File: fig07_user_interface_labelled.pdf (+ .png) · Style: direct labelling

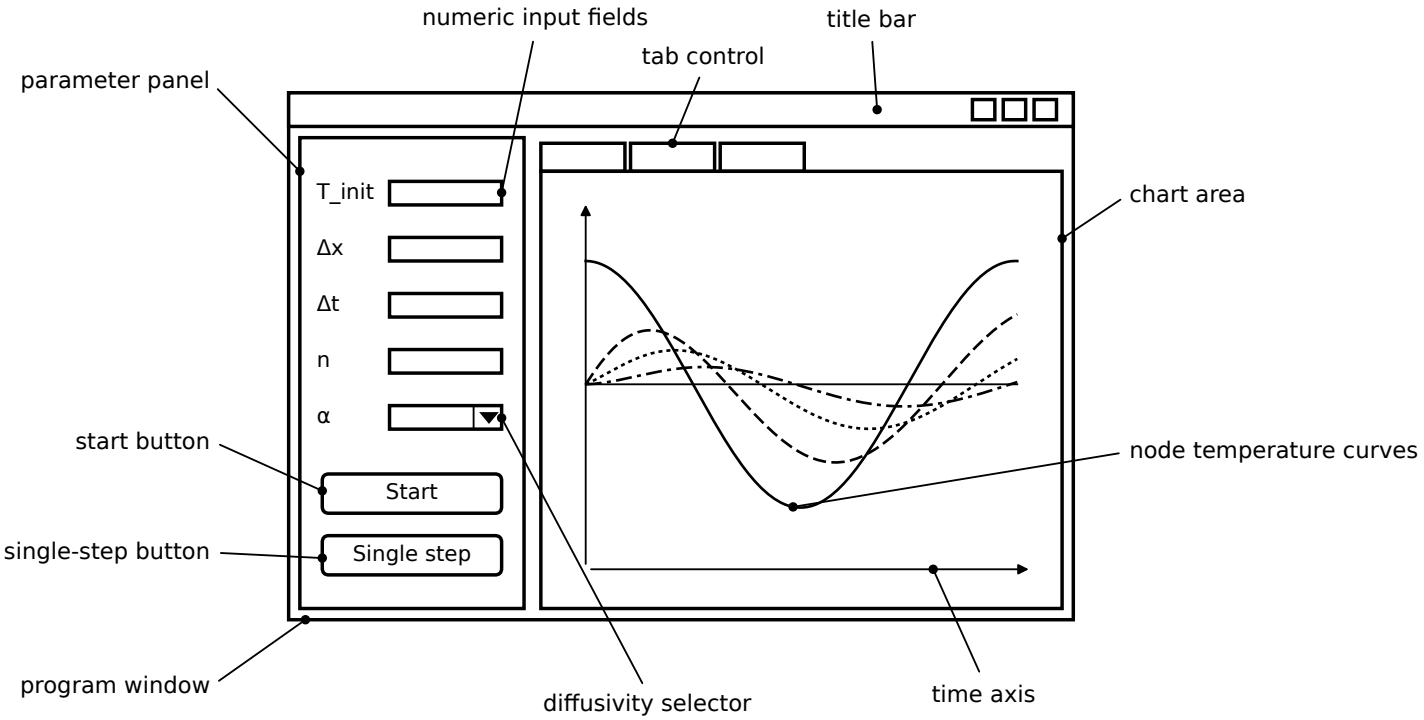


FIG. 7

Direct-labelling variant of the drawing on the previous page: same view and part layout, with the reference numerals replaced by plain-text labels on straight leader lines.